

PhD Qualification Exam in Applied Mathematics

Mock Examination No. 2

Total: 100 points

Problems 1–5 are required. Solve exactly one of Problems 6 and 7. If both are attempted, only the higher score will be counted.

1. (15 points) Let $X_1, \dots, X_N$ be independent uniform random variables on $[0, 1]$. To estimate $I = \int_0^1 e^x dx$, consider the control-variate estimator

$$\hat{I}_c = \frac{1}{N} \sum_{i=1}^N \left[e^{X_i} - c \left(X_i - \frac{1}{2} \right) \right].$$

- (a) Prove that the estimator is unbiased for every $c \in \mathbb{R}$.
- (b) Determine the variance-minimizing coefficient c_* explicitly and prove that it gives a strict variance reduction over ordinary Monte Carlo.
- (c) Give a consistent estimator of the variance and an asymptotic $(1 - \alpha)$ confidence interval for I .

2. (15 points) Let $F : \mathbb{R}^n \rightarrow \mathbb{R}^n$, $s_k = x_{k+1} - x_k \neq 0$, and $y_k = F(x_{k+1}) - F(x_k)$. A new Jacobian approximation is required to satisfy $B_{k+1}s_k = y_k$.

- (a) Among all matrices satisfying this secant equation, derive the unique matrix minimizing $\|B - B_k\|_F$.
- (b) Verify the secant equation, state the resulting quasi-Newton iteration, and show how the update reduces to the scalar secant method when $n = 1$.
- (c) State the arithmetic cost of the rank-one update and identify the numerical breakdown that must be monitored.

3. (20 points) Suppose $A \in \mathbb{R}^{m \times n}$, $m \geq n$, has full column rank and a thin QR factorization $A = QR$. A new observation $a^T x \approx \beta$ is appended to the least-squares problem $Ax \approx b$.

- (a) Reduce the augmented problem to one involving $\begin{bmatrix} R \\ a^T \end{bmatrix}$ and $\begin{bmatrix} Q^T b \\ \beta \end{bmatrix}$, keeping track of the old orthogonal residual.
- (b) Describe a Givens-rotation update that produces an upper triangular $\tilde{R}$ and the updated minimizer. Prove that the method solves the augmented least-squares problem.
- (c) Give the operation count and storage order, and compare them with recomputing a QR factorization from scratch.

4. (15 points) Let $A = A^T > 0$ and let $M = M^T > 0$ be a preconditioner for $Ax = b$.

- (a) Show that the preconditioned problem is equivalent to an SPD system with matrix $C = M^{-1/2} A M^{-1/2}$.
- (b) Derive the standard preconditioned conjugate-gradient recurrences using residuals r_k , preconditioned residuals $z_k = M^{-1} r_k$, and search directions p_k .
- (c) Prove or clearly derive the A -norm estimate

$$\|x_k - x_*\|_A \leq 2 \left(\frac{\sqrt{\kappa_2(C)} - 1}{\sqrt{\kappa_2(C)} + 1} \right)^k \|x_0 - x_*\|_A.$$

Explain the work required per iteration and why M must preserve positive definiteness.

5. (15 points) Let A_x and A_y be the standard second-difference matrices in the two coordinate directions for the two-dimensional heat equation on a rectangular tensor grid. Consider

$$(I - \frac{\tau}{2}A_x)(I - \frac{\tau}{2}A_y)U^{n+1} = (I + \frac{\tau}{2}A_x)(I + \frac{\tau}{2}A_y)U^n.$$

Assume periodic boundary conditions, or homogeneous Dirichlet conditions for which the tensor-product operators commute. Determine the temporal and spatial orders. Prove unconditional ℓ^2 stability by Fourier or eigenmode analysis, and describe an implementation using one-dimensional solves.

6. (20 points; choose one of Problems 6–7)

- (a) Derive the $[1/1]$ Padé approximant to e^z by matching its Taylor series through z^2 .
- (b) Show that the resulting rational function is the stability function of the trapezoidal method for $y' = f(t, y)$, and prove that the method has order two.
- (c) Prove that the method is A -stable but not L -stable. Explain the consequence for very stiff decaying modes.
- (d) Derive the $[0/1]$ Padé approximant, identify the corresponding ODE method, and compare its stiff-decay behavior with the trapezoidal method.

7. (20 points; choose one of Problems 6–7) Let $K \subset \mathbb{R}^n$ be nonempty, closed, and convex, and let $f(x) = \frac{1}{2}x^T Ax - b^T x$ with $A = A^T > 0$. Consider projected gradient

$$x_{k+1} = \Pi_K(x_k - \alpha(Ax_k - b)).$$

- (a) Prove that x_* minimizes f over K if and only if it is a fixed point of this iteration for any $\alpha > 0$.
- (b) Using nonexpansiveness of the projection, prove linear convergence for $0 < \alpha < 2/\lambda_{\max}(A)$.
- (c) Determine the constant step minimizing the resulting worst-case contraction factor, and express the factor in terms of $\kappa_2(A)$.